

Department of Mathematics, IIT Madras
MA 2031 - Linear Algebra for Engineers
End-semester Examination
PART-A

Date: December 14, 2020. Time: 70 minutes Max. marks: 30

Answer all questions. Each question carries 5 marks.

1. Let V be a vector space over $\mathbb{F}$, $w \in V$ and $\{v_1, v_2, \dots, v_n\}$ be a linearly independent subset of V . Prove that if $\{v_1 + w, v_2 + w, \dots, v_n + w\}$ is a linearly dependent set, then $w \in \text{Span}\{v_1, v_2, \dots, v_n\}$. Is the converse true? Justify.
2. Suppose that P, Q and R are nontrivial subspaces of a vector space V such that $P + Q = P + R$. Should $Q = R$ necessarily? Justify your conclusion.
3. Let S denote the subset $\{(1, -1, 0), (0, -3, 2), (2, 1, -2), (3, 3, -4)\}$ of $\mathbb{R}^3$. Find the smallest set $R \subset S$ such that $\text{Span}(R) = \text{Span}(S)$.
4. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, where a, b, c, d are real numbers. For $x, y \in \mathbb{R}^2$, let $f_A(x, y) = y^T A x$ be an inner product on $\mathbb{R}^2$. Show that $b = c$, $a > 0$, $d > 0$, and $ad - bc > 0$.
5. Let V and W be finite dimensional vector spaces over $\mathbb{F}$ and $T : V \rightarrow W$ is a linear map. When is T said to be an isomorphism? Prove that T is an isomorphism if and only if $\{T(v_1), T(v_2), \dots, T(v_n)\}$ is a basis for W whenever $\{v_1, v_2, \dots, v_n\}$ is a basis for V .
6. Let V and W be finite dimensional inner product spaces over the same field. Let $T : V \rightarrow W$ and $S : W \rightarrow V$ be linear transformations. Suppose that $TS = 0$. Then show the following:
 - (a) $\dim V \geq \text{rank}(T) + \text{rank}(S)$.
 - (b) $\dim W \leq \text{rank}(T^*) + \text{rank}(S^*)$.
